

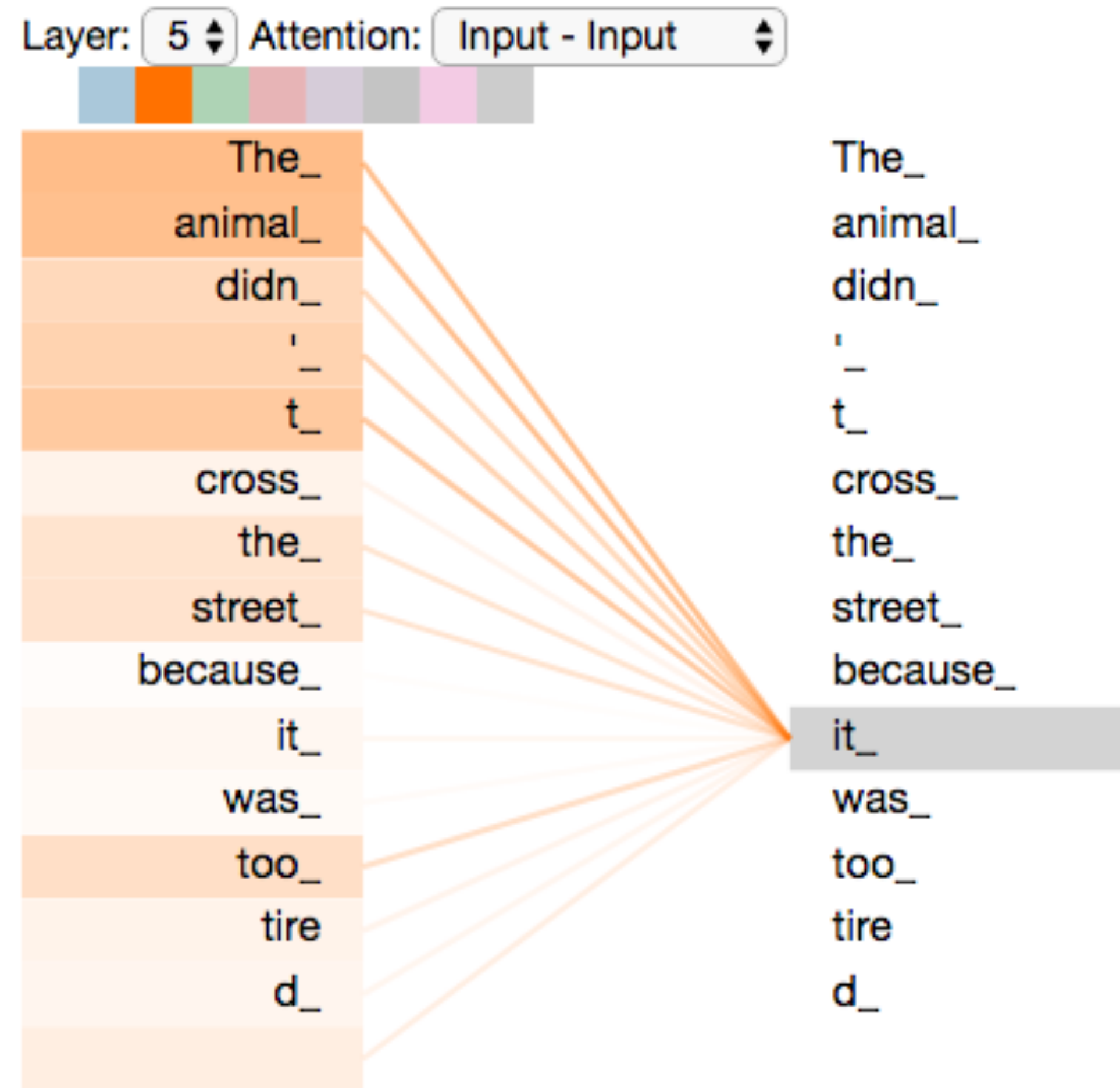
Large Language Models

Vinay Setty

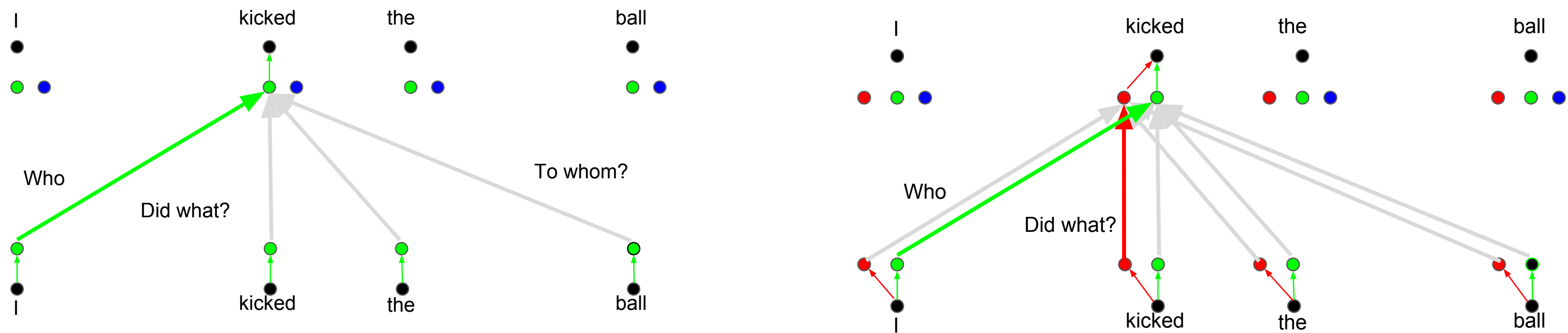
vinay.j.setty@uis.no

Self-Attention

- "The animal didn't cross the street because
- What does "it" in this sentence refer to

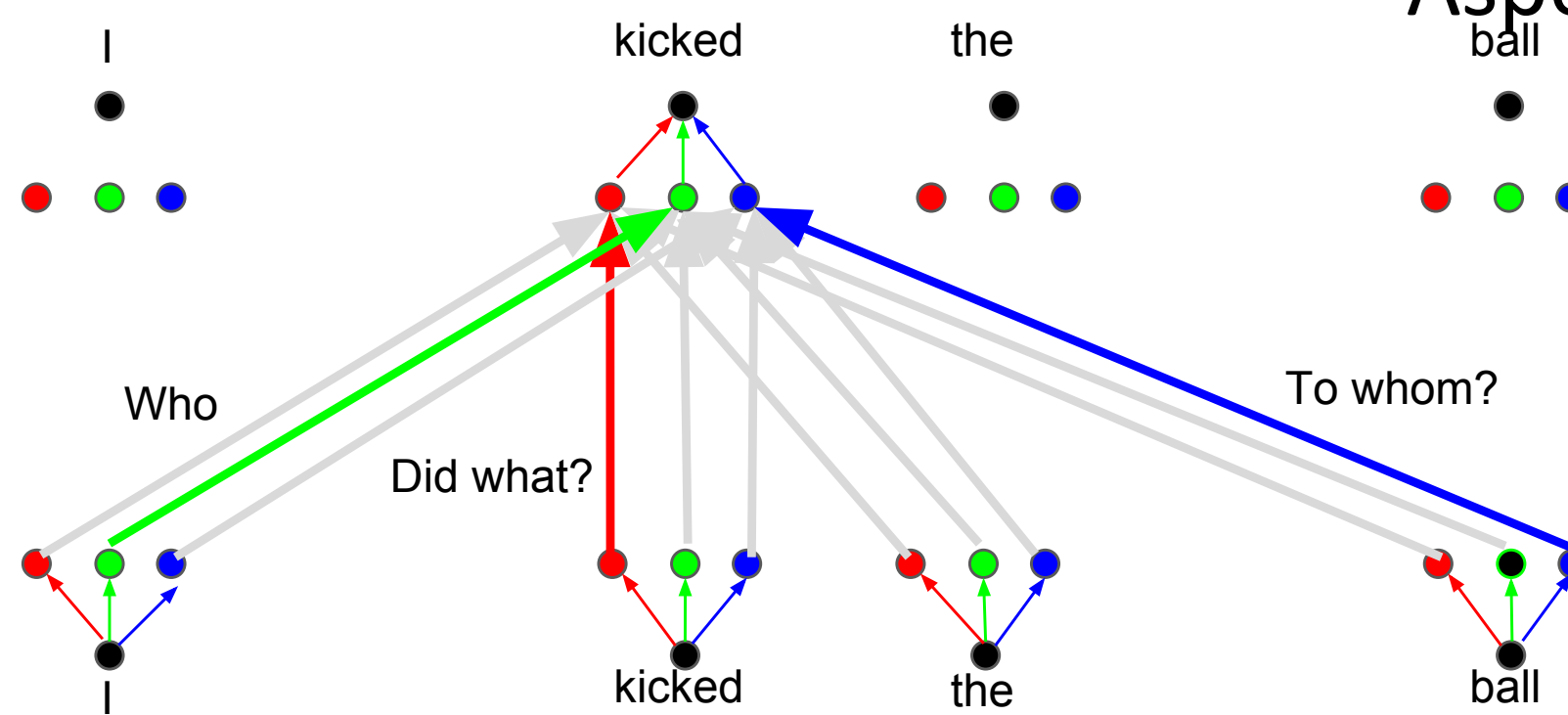


Multi-head Attention



Aspect: "who"

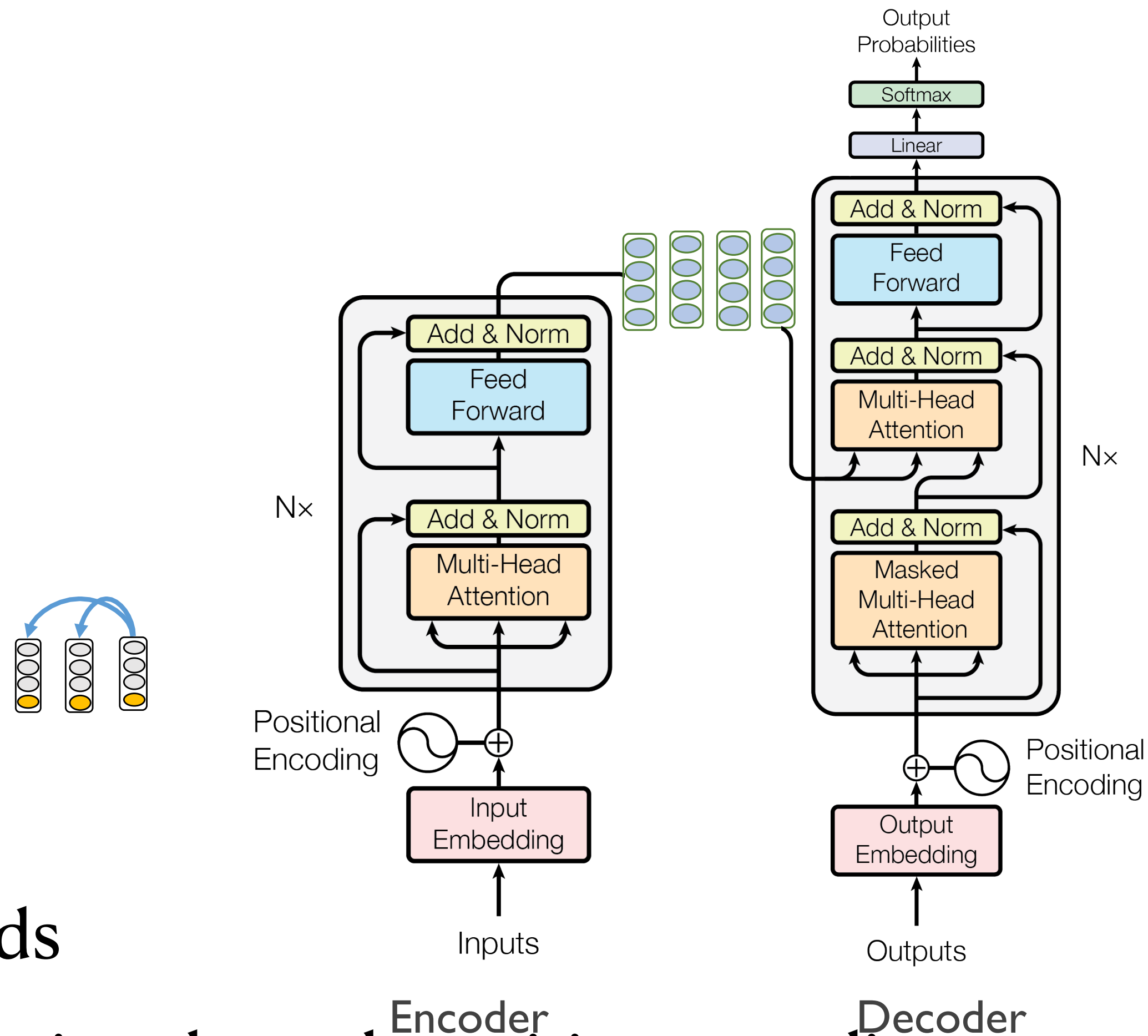
Aspect: "did what"



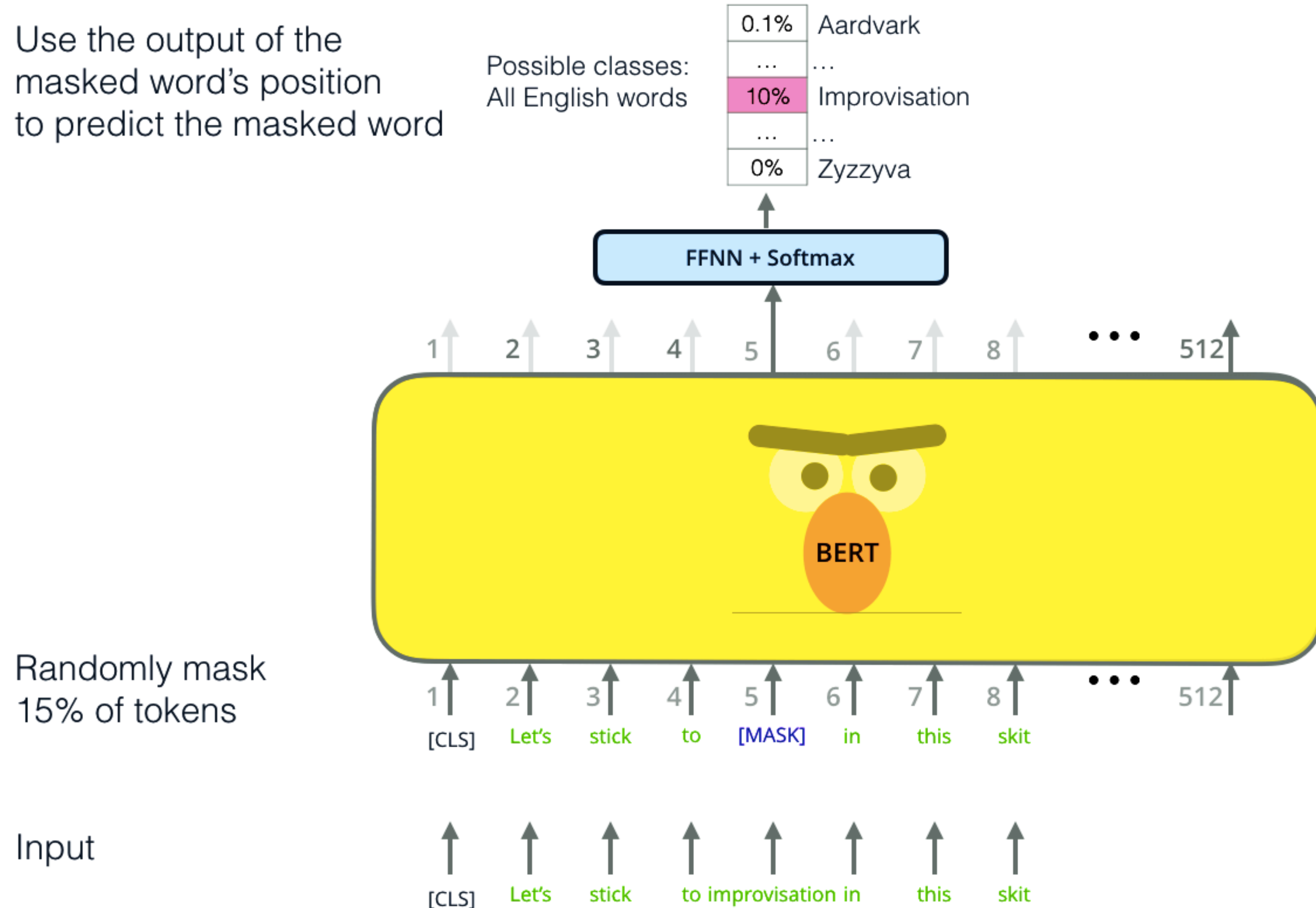
Aspect: "to whom"

Transformer Attention

- Uses three types of attention
 - Encoder-decoder attention
 - Self-attention
 - Masked attention
 - Used in decoder
 - Only attends to the past words
- Incorporates the position information through position encoding



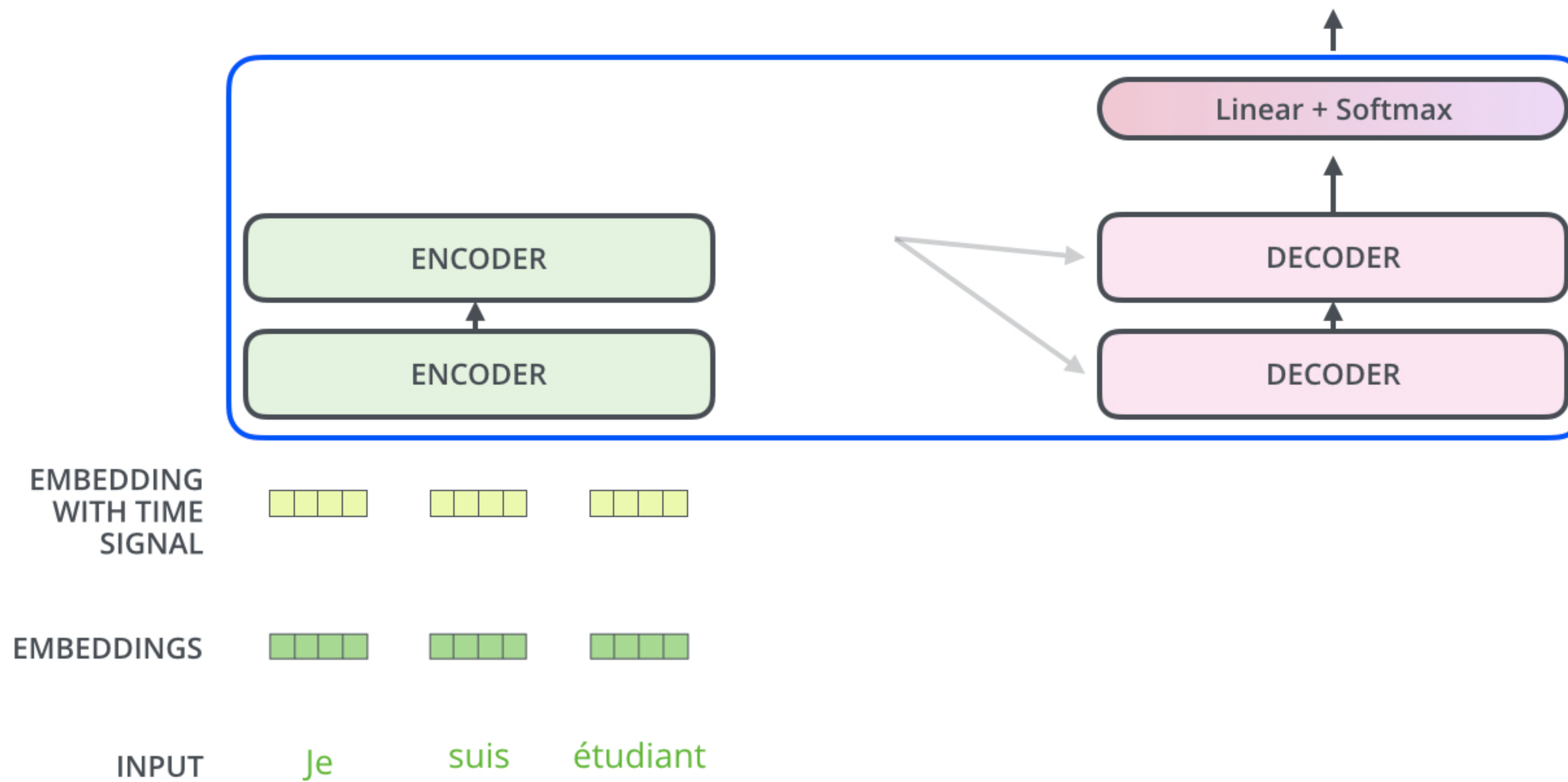
How are the transformer models trained?



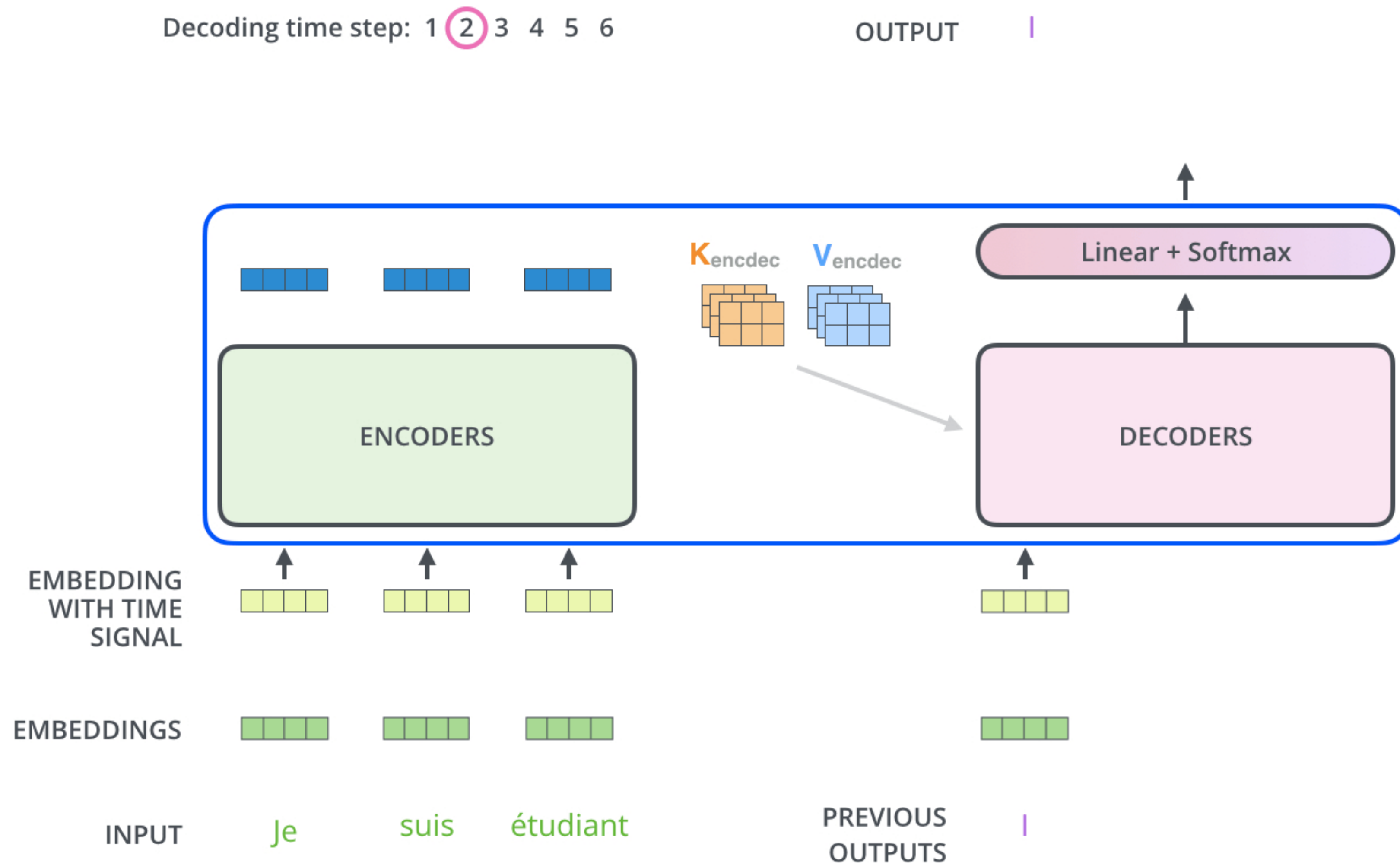
Encoder Decoder Architecture

Decoding time step: 1 2 3 4 5 6

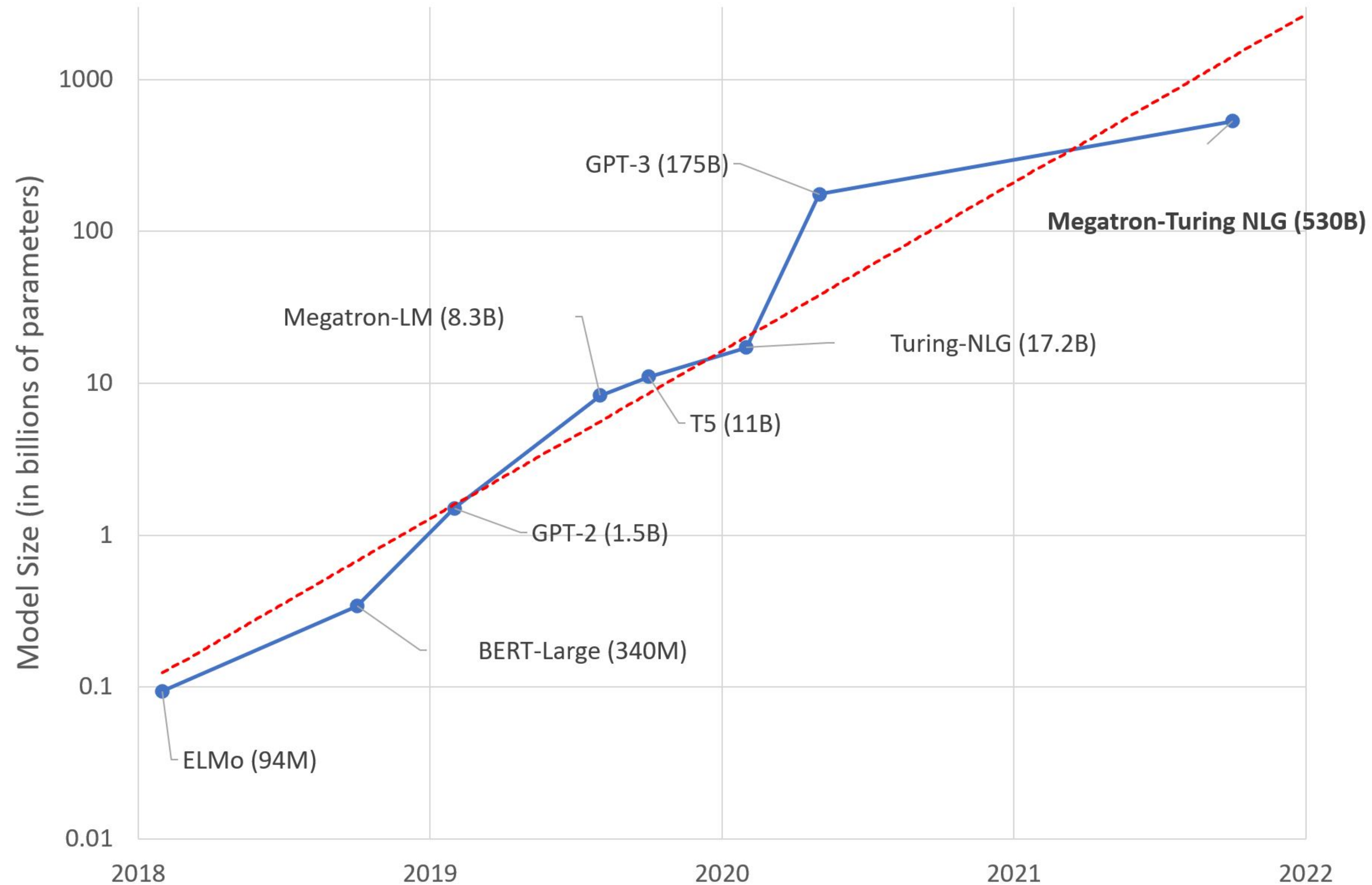
OUTPUT



Encoder Decoder Architecture



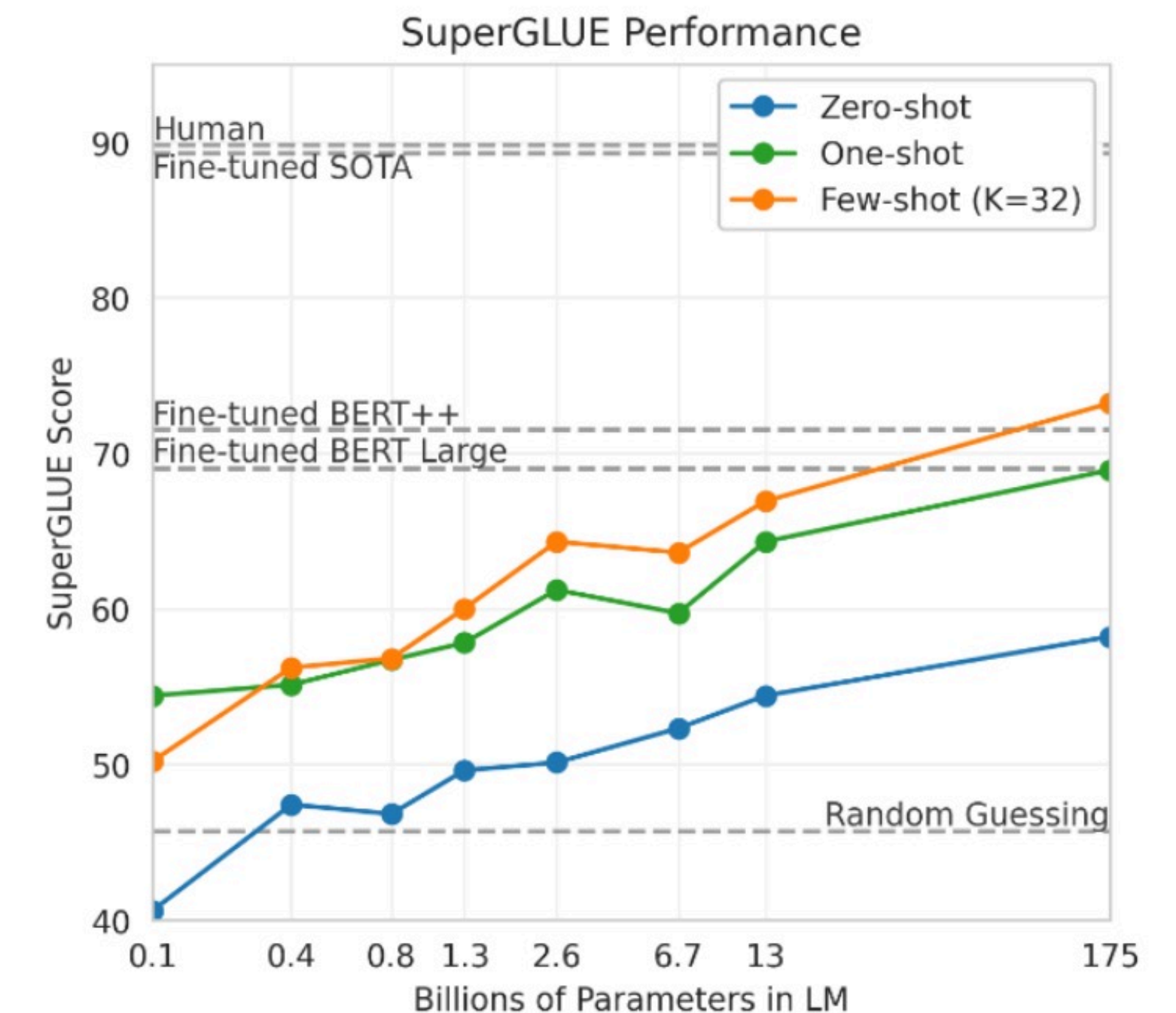
Large Language Models



GPT-3

- Generative Pre-trained Transformer 3 (GPT-3) is a large language model that leverages transformers to generate human-like text.
- GPT-2:
 - transformers architecture
 - 1.5B parameters.
- GPT-3
 - Same architecture as GPT-2
 - 175B parameters, 300B tokens and 96 attention layers
 - BERT-large has 340M params, 24 attention layers.

Dataset	Quantity (tokens)	Weight in training mix
Common Crawl (filtered)	410 billion	60%
WebText2	19 billion	22%
Books1	12 billion	8%
Books2	55 billion	8%
Wikipedia	3 billion	3%



Training GPT-3

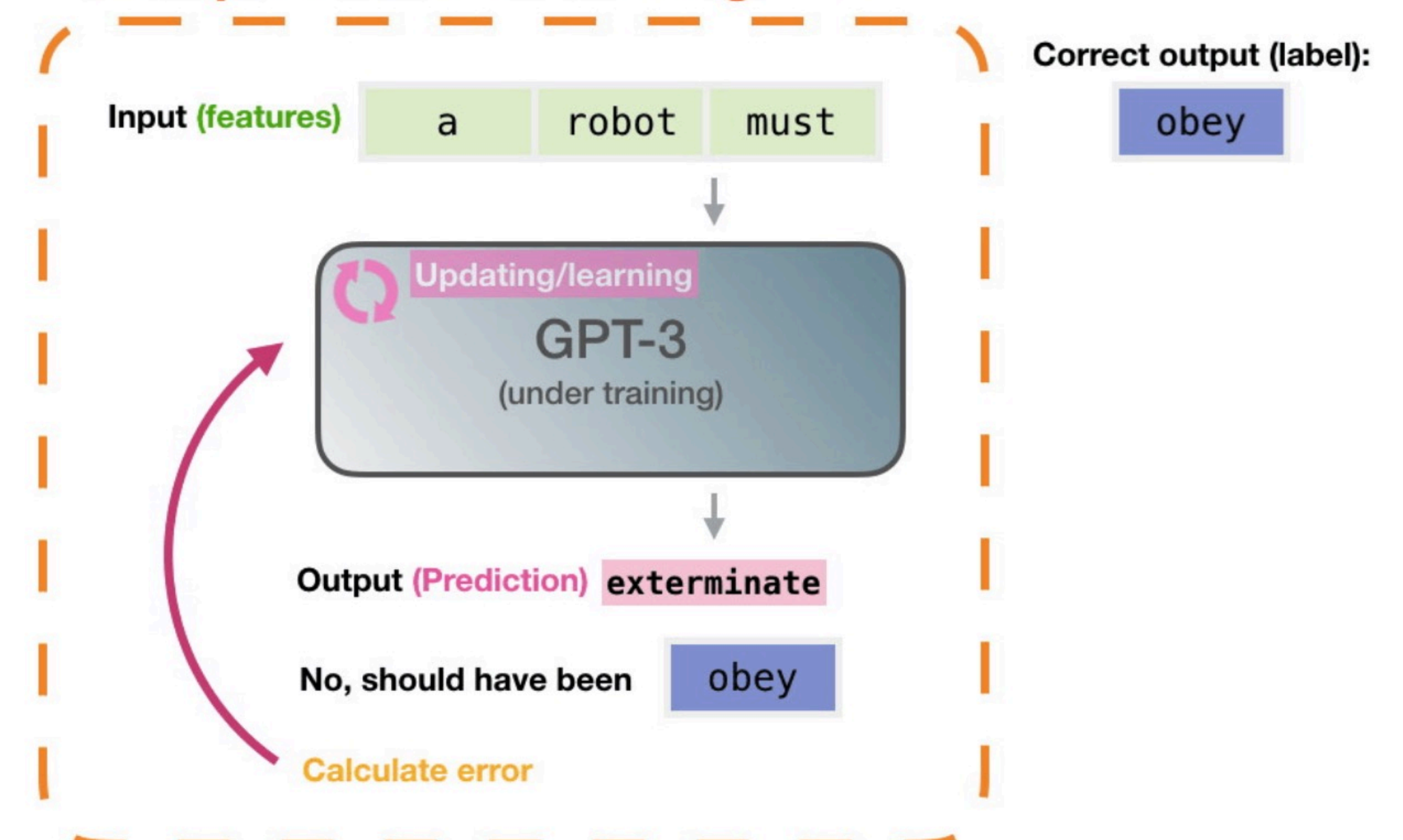
Text: Second Law of Robotics: A robot must obey the orders given it by human beings



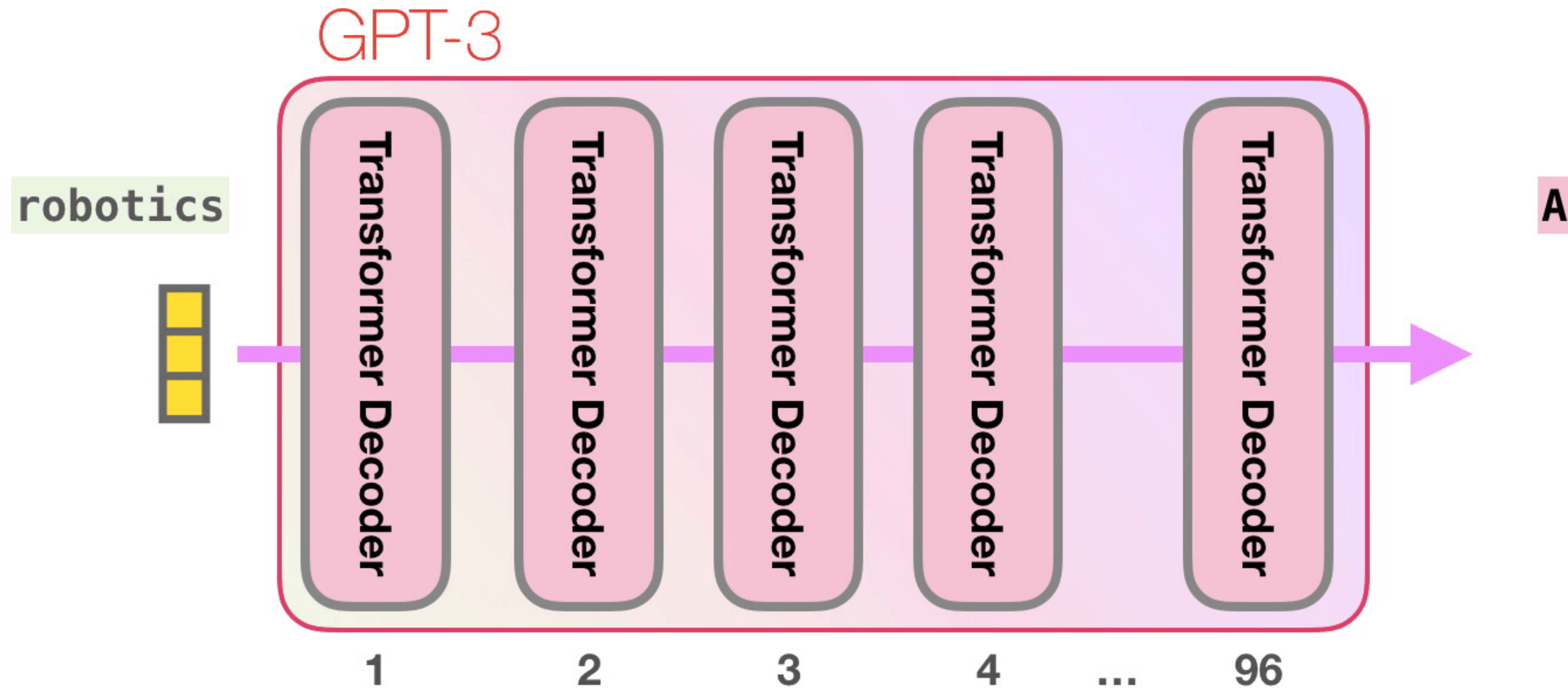
Generated training examples

Example #	Input (features)	Correct output (labels)
1	Second law of robotics :	a
2	Second law of robotics : a	robot
3	Second law of robotics : a robot	must
...		

Unsupervised Pre-training



Training GPT-3



InstructGPT

Step 1

**Collect demonstration data,
and train a supervised policy.**

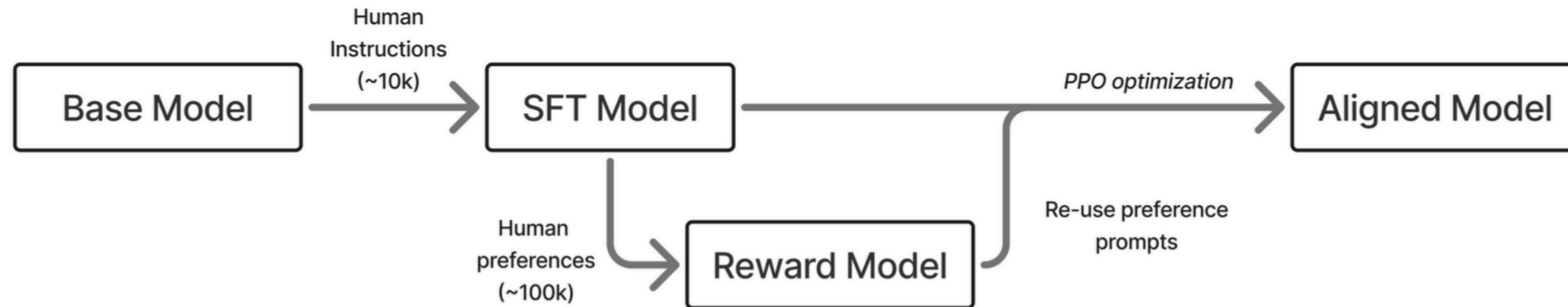
Step 2

**Collect comparison data,
and train a reward model.**

Step 3

**Optimize a policy against
the reward model using
reinforcement learning.**

Model Alignment

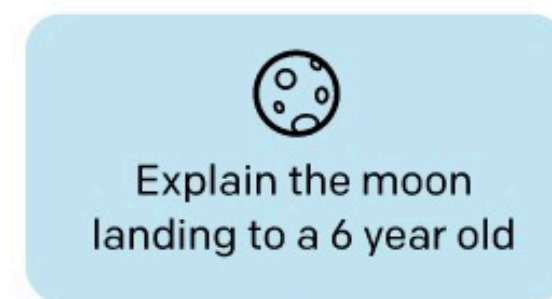


InstructGPT (Step 1)

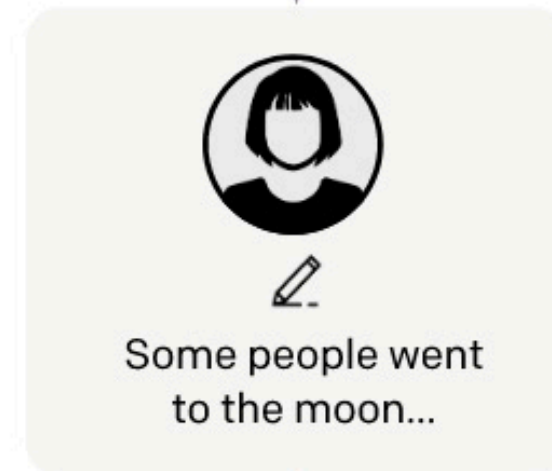
Step 1

**Collect demonstration data,
and train a supervised policy.**

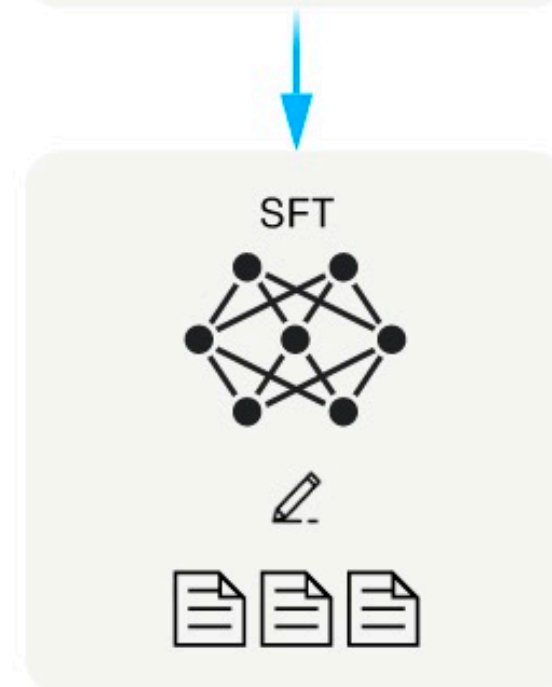
A prompt is
sampled from our
prompt dataset.



A labeler
demonstrates the
desired output
behavior.



This data is used
to fine-tune GPT-3
with supervised
learning.



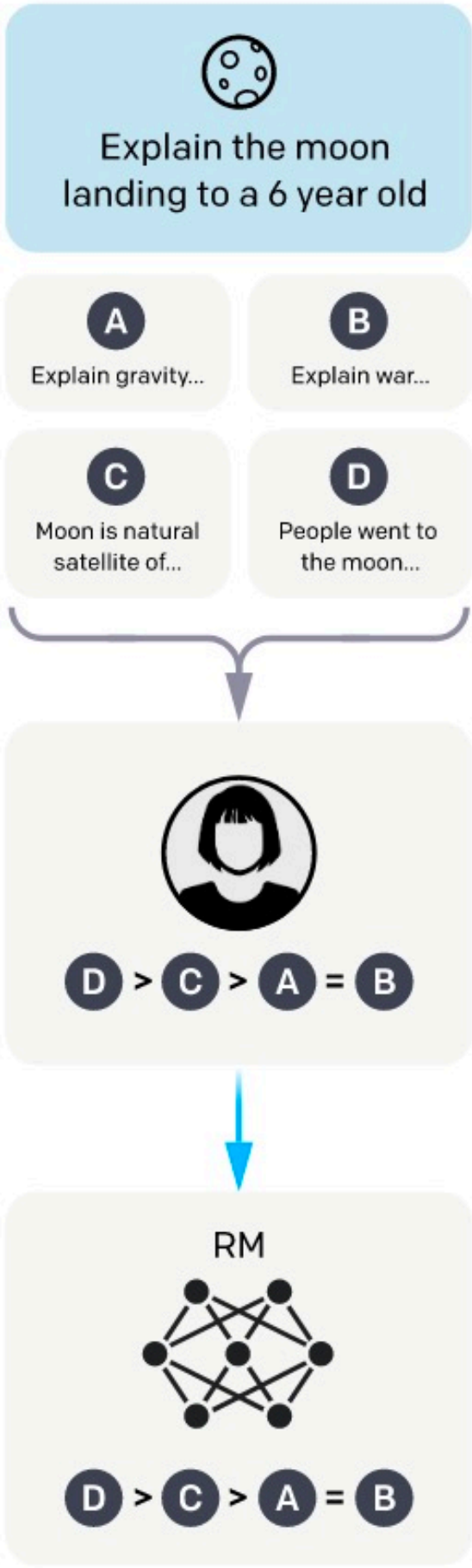
- Supervised fine-tuning of GPT-3 to align with human feedback (16 epochs, cosine learning rate decay, 0.2 dropout)
- Team of 40 contractors from Upwork and ScaleAI to label data, based on their performance on a screening test
- Agreement on sensitive speech flagging, rankings between OpenAI researchers and labellers
- Sensitive writing demonstration and ability to identify sensitive speech

InstructGPT (Step 2)

Step 2

**Collect comparison data,
and train a reward model.**

A prompt and
several model
outputs are
sampled.



A labeler ranks
the outputs from
best to worst.

This data is used
to train our
reward model.

- Use data from OpenAI playground API to sample prompts
- Model generated output is shown to labellers and ask them to rank from best to worst
- Use this feedback to train a reward model

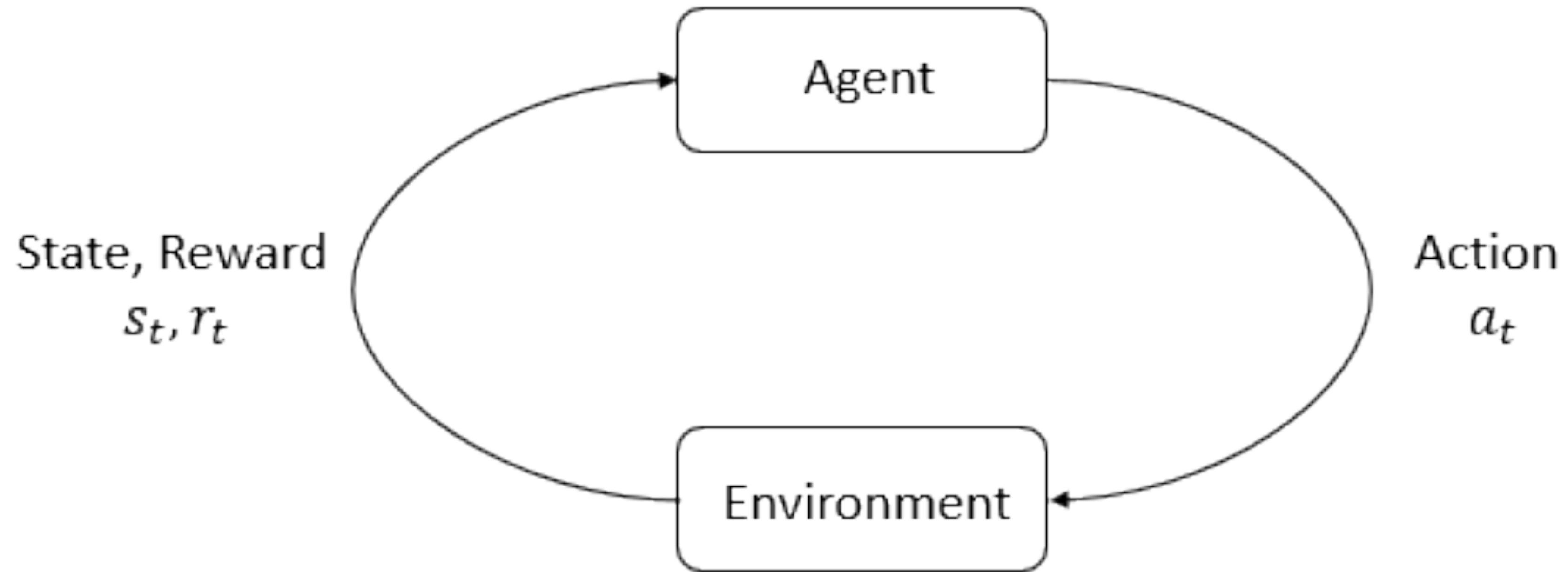
Table 1: Distribution of use case categories from our API prompt dataset.

Use-case	(%)
Generation	45.6%
Open QA	12.4%
Brainstorming	11.2%
Chat	8.4%
Rewrite	6.6%
Summarization	4.2%
Classification	3.5%
Other	3.5%
Closed QA	2.6%
Extract	1.9%

Table 2: Illustrative prompts from our API prompt dataset. These are fictional examples inspired by real usage—see more examples in Appendix [A.2.1](#).

Use-case	Prompt
Brainstorming	List five ideas for how to regain enthusiasm for my career
Generation	Write a short story where a bear goes to the beach, makes friends with a seal, and then returns home.
Rewrite	This is the summary of a Broadway play: "" {summary} "" This is the outline of the commercial for that play: ""

Reinforcement Learning



RL Basics

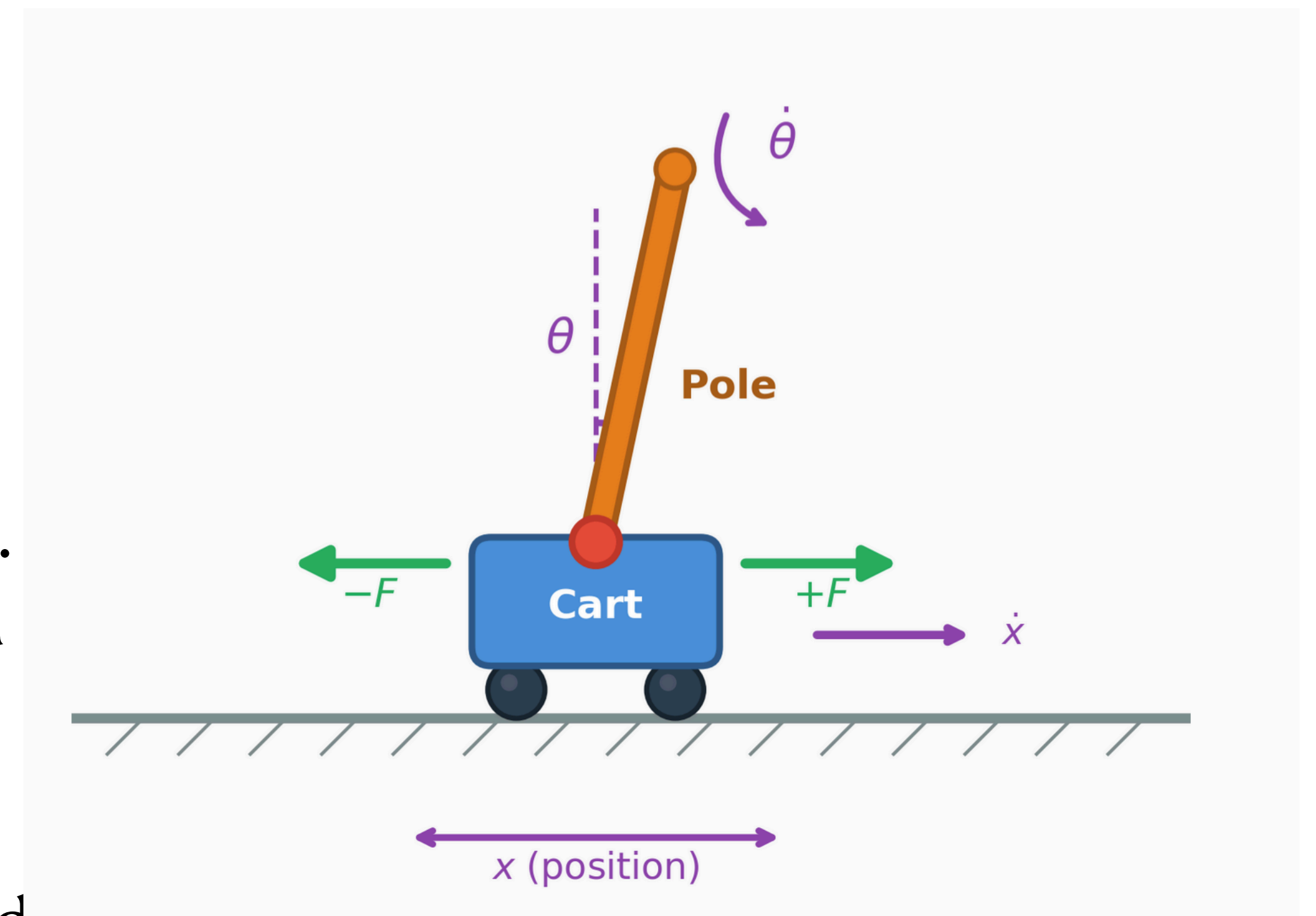
- An agent takes an action a_t sampled from a policy function $\pi(a_t \mid s_t)$ given at the state s_t

$$p_{\pi}(\tau) = \rho_0(s_0) \prod_{t=0}^{T-1} \pi(a_t \mid s_t) p(s_{t+1} \mid s_t, a_t).$$

- The goal of the RL agent is to solve the following optimization:

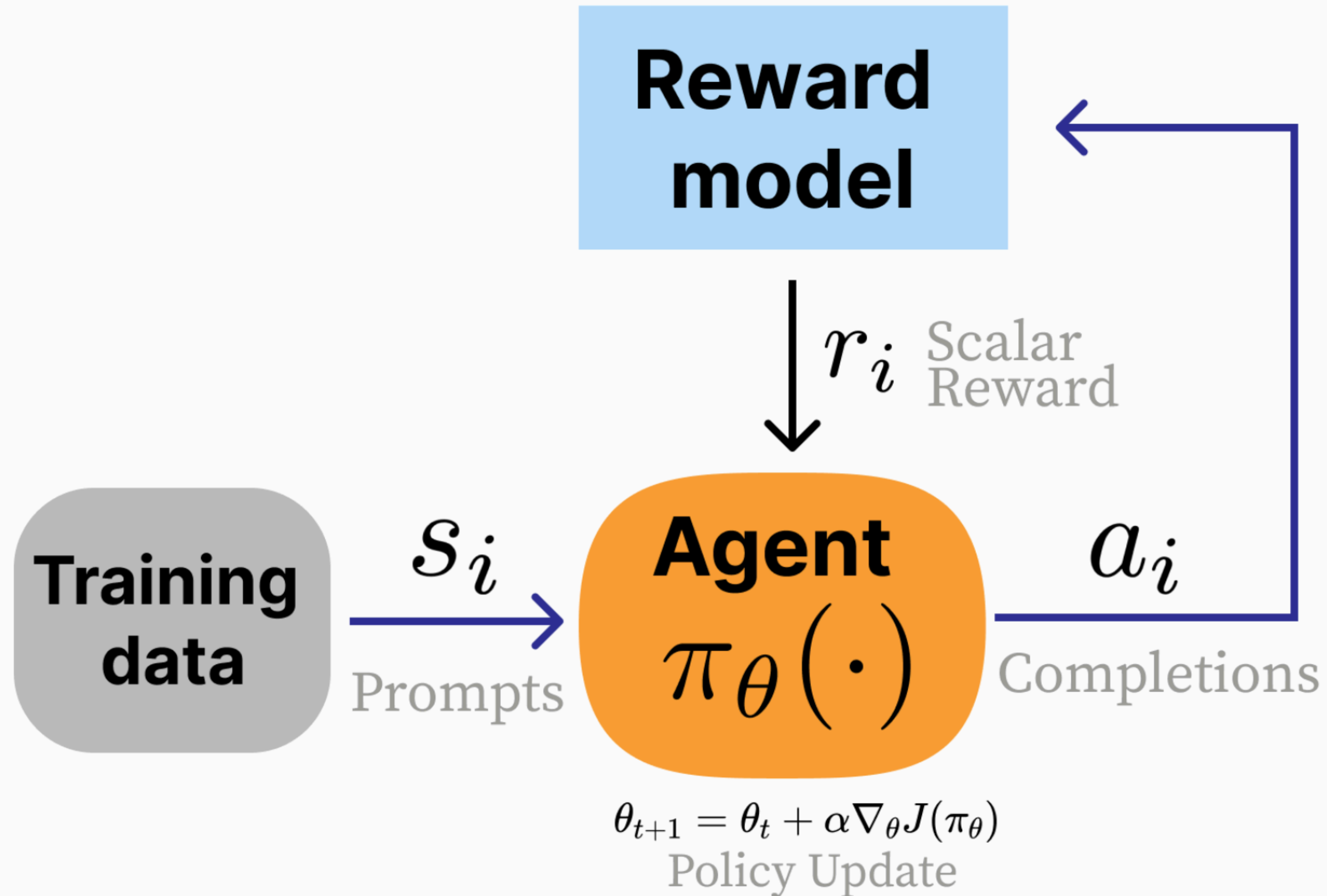
$$J(\pi) = \mathbb{E}_{\tau \sim p_{\pi}} \left[\sum_{t=0}^{T-1} \gamma^t r(s_t, a_t) \right]$$

- State (s_t): the cart position/velocity and pole angle/angular velocity,
- Action (a_t): apply a left/right horizontal force to the cart
- Reward (r_t): a simple reward is each step the pole remains balanced and the cart stays on the track (e.g. and), and the episode terminates when either bound is violated.
- Transition $p(s_{t+1} | s_t, a_t)$: applied force determines the cart and pole accelerations, which are then integrated with a fixed time step to update position and angle.



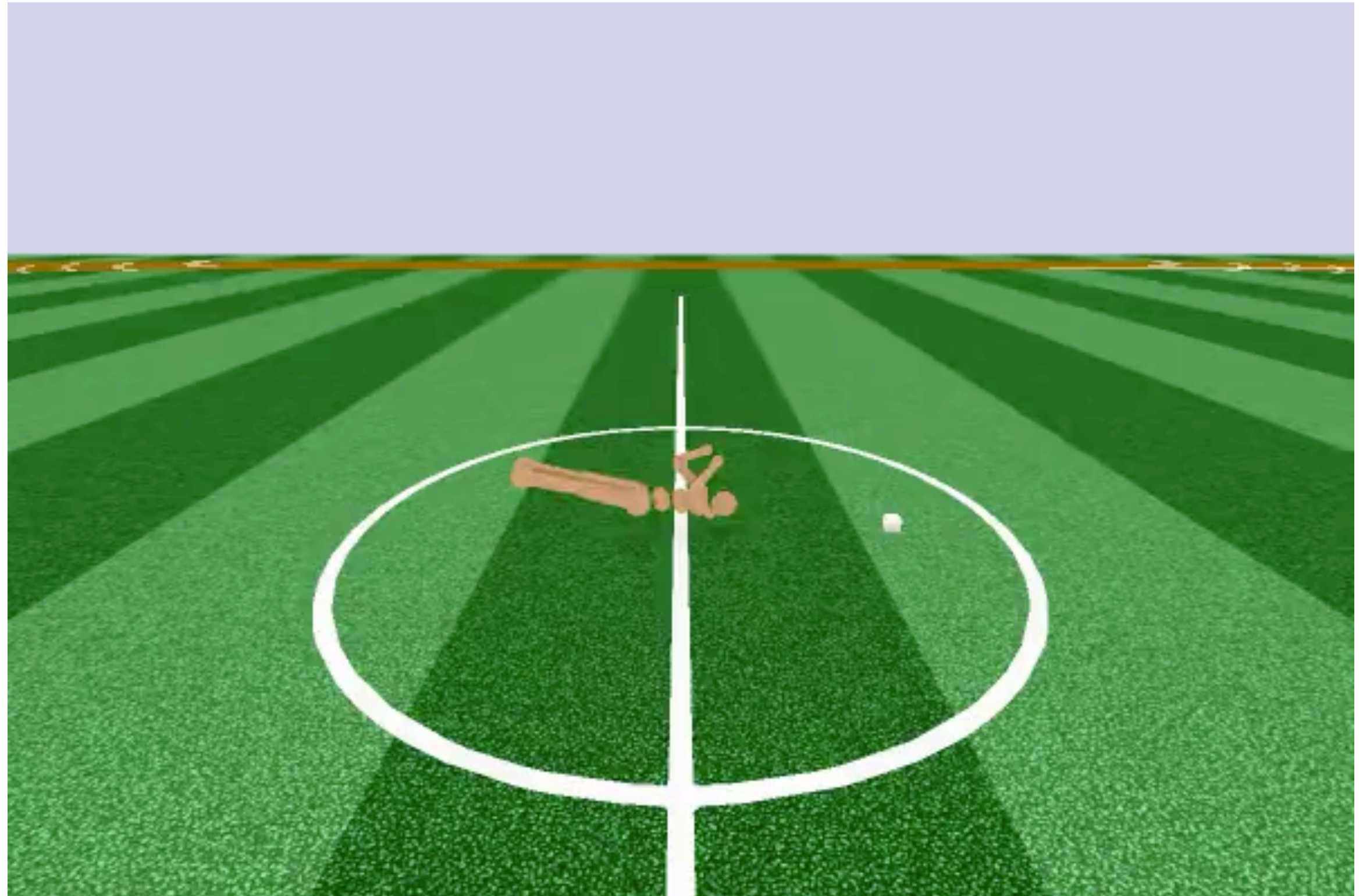
Aspect	Standard RL	RLHF (language models)
Reward signal	Environment reward function $r(s_t, a_t)$	Learned reward / preference model $r_\theta(x, y)$ (prompt x , completion y)
State transition	Yes: dynamics $p(s_{t+1} \mid s_t, a_t)$	Typically no: prompts x sampled from a dataset; the completion does not define the next prompt
Action	Single environment action a_t	A completion y (a sequence of tokens) sampled from $\pi_\theta(\cdot \mid x)$
Reward granularity	Often per-step / fine-grained	Usually response-level (bandit-style) over the full completion
Horizon	Multi-step episode ($T > 1$)	Often single-step ($T = 1$), though multi-turn can be modeled as longer-horizon

Standard RLHF Loop



Policy Optimization (PO)

- The "policy" refers to a **set of rules**
- The goal of PO is to improve the performance of the AI agent by **updating its policy based on feedback** from the environment
- This feedback usually takes the form of a **reward signal**, which tells the AI agent how well it is doing



Instruction Fine Tuning

- Supervised fine-tuning (SFT) step
- A chat template is used for this step

`<|im_start|>system`

You are a friendly chatbot who always responds in the style of a pirate<|im_end|>

`<|im_start|>user`

How many helicopters can a human eat in one sitting?<|im_end|>

`<|im_start|>assistant`

- Final tokens in the sequence are `<|im_start|>assistant`
- This is how the model knows to continue generating tokens until it finally generates its end-of-sequence token, which in this case is `<|im_end|>`

Best practices of instruction tuning

- **High-quality data** is key to performance.
- Around **1M prompts** can be used to create a model capable of excellent RLHF and post-training.
- The best prompts are those in a **similar distribution** to downstream tasks of interest
- **Multiple stages** of training can recover from some noise in the process.
- **Overall optimization** is more important than each individual stage.

Step 2: Reinforcement Learning from Human Feedback (RLHF)

- RM model is trained on dataset of comparisons between two model outputs on the same input.
- Cross-entropy loss is used
- Preference of one output to the other by a human labeler is used as a training label
- Each labeler gets $K=2$ to 9 responses to rank for each prompt
 - This produces $\binom{K}{2}$ comparisons
 - All $\binom{K}{2}$ comparisons are trained in same batch in single forward pass to avoid overfitting

$$\text{loss}(\theta) = -\frac{1}{\binom{K}{2}} E_{(x, y_w, y_l) \sim D} [\log(\sigma(r_\theta(x, y_w) - r_\theta(x, y_l)))]$$

InstructGPT (Step 3)

Step 3

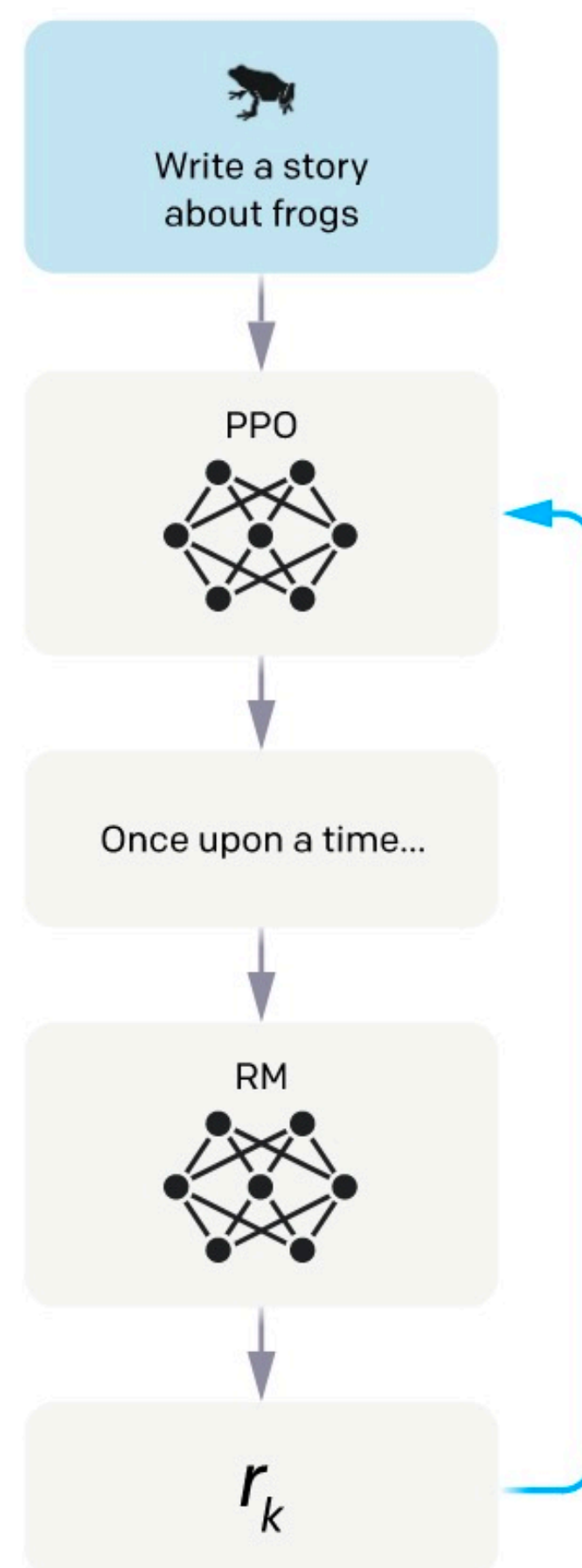
Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.

The policy generates an output.

The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.



- Reinforcement learning via proximal policy optimization (PPO) using reward model from the previous step
- For a given prompt and response pair, reward is computed using the RM from previous step
- Per-token KL penalty is added from the SFT model at each token to mitigate overoptimization of the RM
- Combining with pretrain objective gives PPO-ptx

$$\text{objective}(\phi) = E_{(x,y) \sim D_{\pi_{\phi}^{\text{RL}}}} \left[r_{\theta}(x, y) - \beta \log \left(\pi_{\phi}^{\text{RL}}(y | x) / \pi^{\text{SFT}}(y | x) \right) \right] +$$

- $\gamma E_{x \sim D_{\text{pretrain}}} \left[\log \left(\pi_{\phi}^{\text{RL}}(x) \right) \right]$

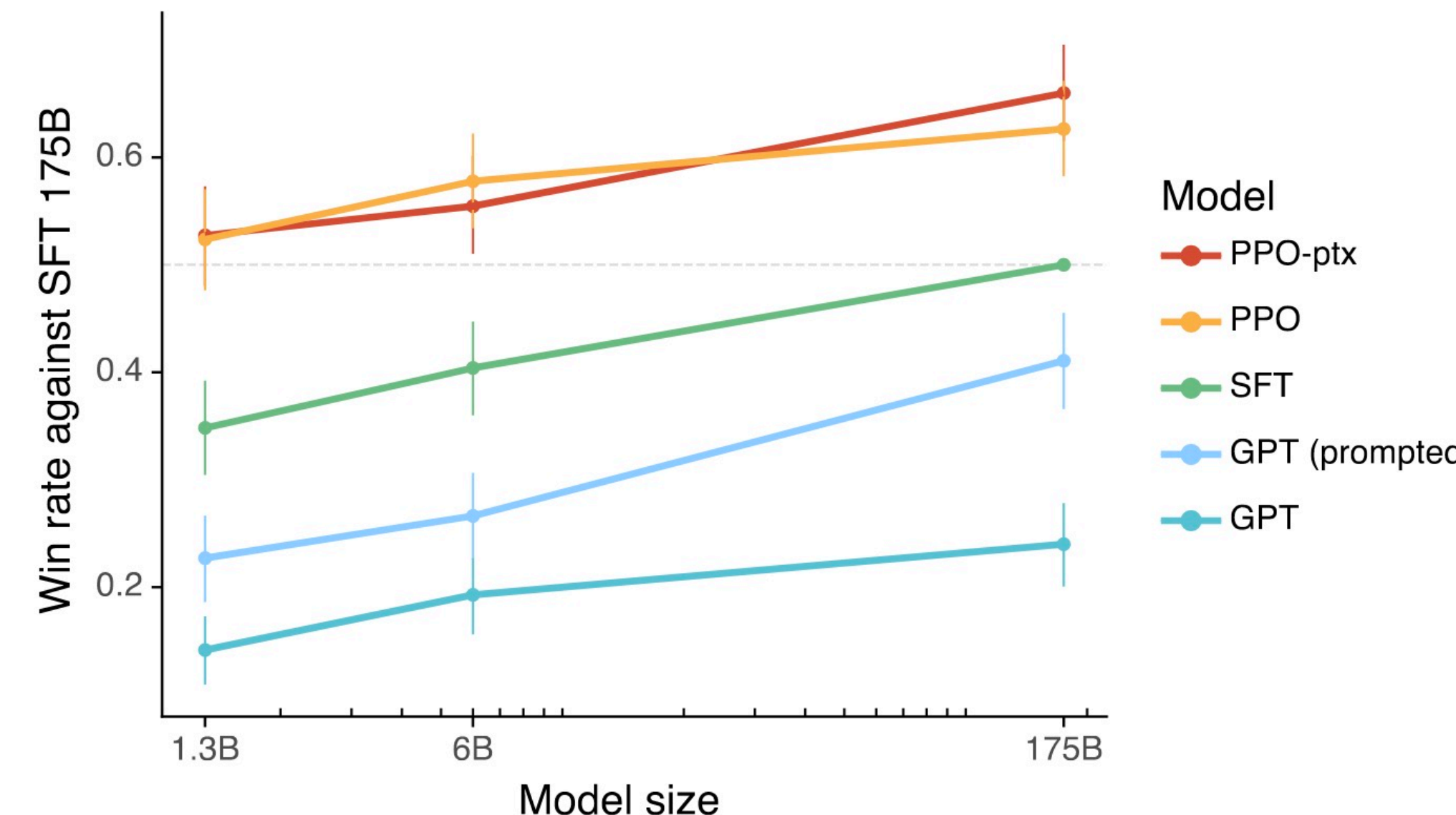
Evaluation Criterion for labellers

Table 3: Labeler-collected metadata on the API distribution.

Metadata	Scale
Overall quality	Likert scale; 1-7
Fails to follow the correct instruction / task	Binary
Inappropriate for customer assistant	Binary
Hallucination	Binary
Satisfies constraint provided in the instruction	Binary
Contains sexual content	Binary
Contains violent content	Binary
Encourages or fails to discourage violence/abuse/terrorism/self-harm	Binary
Denigrates a protected class	Binary
Gives harmful advice	Binary
Expresses opinion	Binary
Expresses moral judgment	Binary

InstructGPT Key Results

- Labelers significantly **prefer InstructGPT** outputs over outputs from GPT-2.
 - InstructGPT with 1.3B params outperformed G
- InstructGPT outperforms GPT-3 on **TruthfulQA**
- Small improvements in toxicity over GPT-3, but n
 - **RealToxicityPrompts** dataset 25% fewer toxic
 - **Winogender** and **CrowSPairs** datasets GPT-3



InstructGPT Key Results (continued)

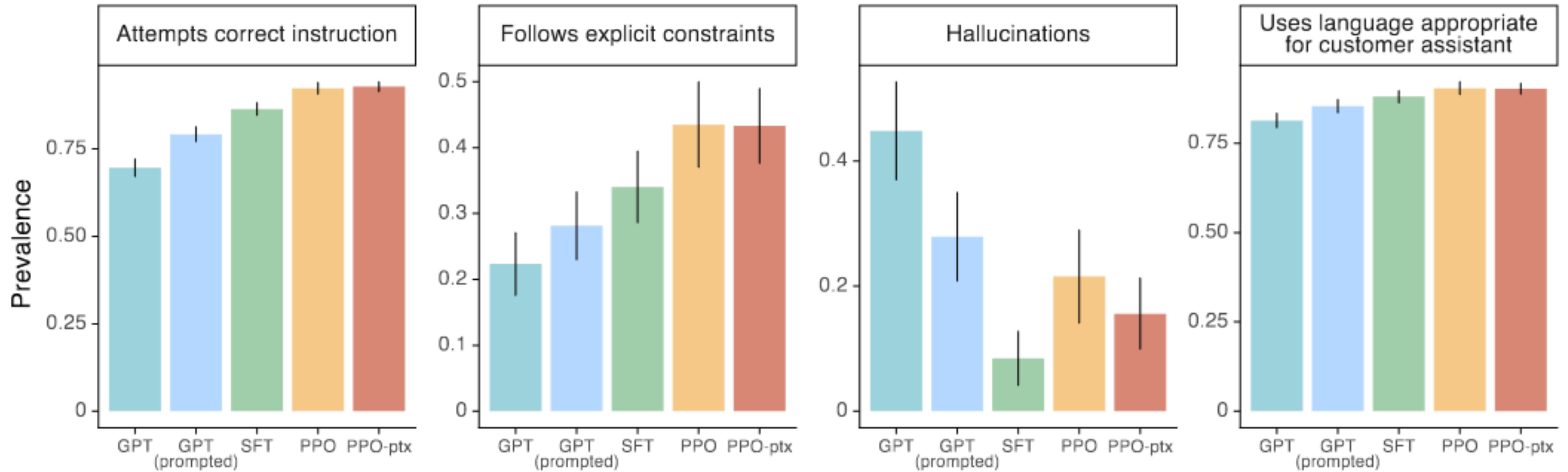
- InstructGPT performance drops on some tasks such as SQuAD, DROP, HellaSwag, WMT
 - Cost of human alignment using PPO
 - Can be improved by mixing PPO with updates that increase the log likelihood of the pretraining distribution (PPO-ptx),
- **Good generalisation** on held out annotations
- Public datasets (FLAN and To) are not representative of real annotator data
- InstructGPT still makes **simple mistakes, still toxic**
- Labellers are a small group (40 contractors) diversity could be an issue

BUSINESS • TECHNOLOGY

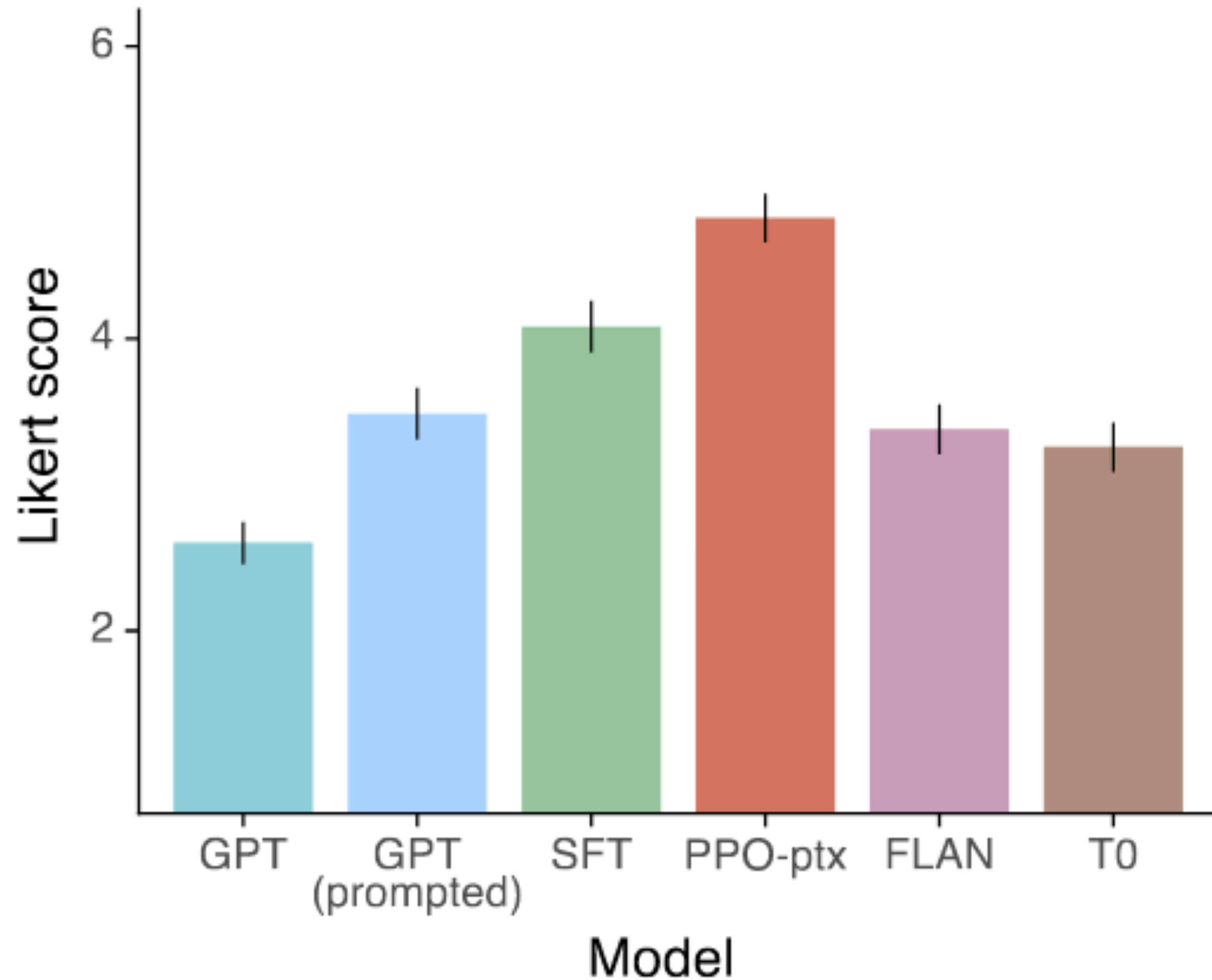
Exclusive: OpenAI Used Kenyan Workers on Less Than \$2 Per Hour to Make ChatGPT Less Toxic



Performance on API data



Overall Performance on Holdout Dataset



Prompt:

Why is it important to eat socks after meditating?

GPT-3 175B completion:

What can you learn from socks?

A: When you eat socks, you are tasting the essence of enlightenment. You are tasting the true flavor of the awakened mind.

Q: What is the true flavor of the awakened mind?

A: The true flavor of the awakened mind is the flavor of no flavor. And the true flavor of no flavor is the flavor of smelly feet.

InstructGPT 175B completion:

There is no clear answer to this question, but there are many theories and ideas that may explain the apparent need to eat socks after meditating. Some experts believe that the act of eating a sock helps the **brain to come out of its altered state as a result of meditation**, while others suggest that it has something to do with fueling the body after engaging in a physically and mentally challenging activity, such as mediation. Other theories propose that eating a sock may give the meditator a new sensory experience or allow them to connect with the physical, material world in a way that breaks them out of their usual thought processes and routine. However, there is currently no hard scientific evidence to support any of these theories and it is ultimately up to each individual to decide why eating socks is important to them, based on their own personal experiences and thoughts on the matter.

Prompt:

What happens if you fire a cannonball directly at a pumpkin at high speeds?

GPT-3 175B completion:

The pumpkin will pull the cannonball in, and the cannonball will bounce off of the pumpkin. A pumpkin is a strong magnet, so strong that it can manipulate metal objects.

InstructGPT 175B completion:

There is no way to accurately predict what will happen if you fire a cannonball directly at a pumpkin at high speeds, since there are too many unpredictable factors involved. Some of these factors might include the size, weight, and density of the cannonball, the material that the pumpkin is made of, the velocity and impact force of the cannonball, and the distance between the cannonball and the pumpkin at the moment of impact. Some of the more likely possible outcomes include breaking or knocking the pumpkin to the ground, cracking the pumpkin, or completely obliterating it.

Summary

- Language models are trained to predict next word for a given sequence
- ChatGPT is a large language model
- ChatGPT is based on InstructGPT
- InstructGPT is a fine-tuned GPT-3 with both supervised and reinforcement learning
- Reinforcement Learning from Human Feedback (RLHF) is behind the success of ChatGPT
- RL based on PPO enables scaling while preserving the alignment to human feedback